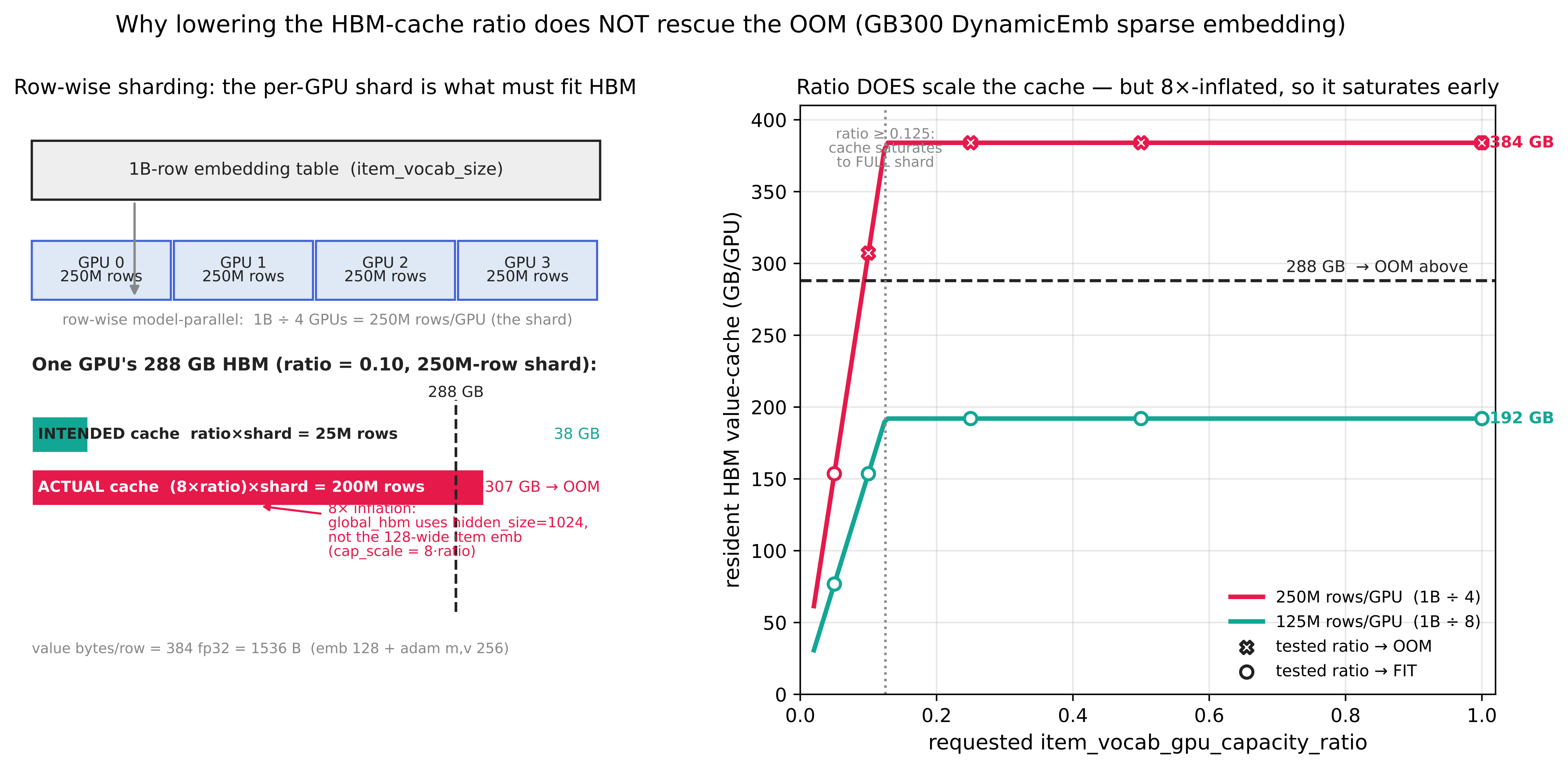
Why lowering the HBM-cache ratio does NOT rescue the OOM (GB300 DynamicEmb sparse embedding)
Row-wise sharding: the per-GPU shard is what must fit HBM
1B-row embedding table (item_vocab_size)
GPU 0 250M rows
GPU 1 250M rows
GPU 2 250M rows
GPU 3 250M rows
row-wise model-parallel: 1B ÷ 4 GPUs = 250M rows/GPU (the shard)
One GPU's 288 GB HBM (ratio = 0.10, 250M-row shard):
288 GB
INTENDED cache ratio×shard = 25M rows
38 GB
ACTUAL cache (8×ratio)×shard = 200M rows
307 GB → OOM
8× inflation: global_hbm uses hidden_size=1024, not the 128-wide item emb (cap_scale = 8·ratio)
value bytes/row = 384 fp32 = 1536 B (emb 128 + adam m,v 256)
Ratio DOES scale the cache — but 8×-inflated, so it saturates early
ratio ≥ 0.125: cache saturates to FULL shard
384 GB
288 GB → OOM above
192 GB
resident HBM value-cache (GB/GPU)
requested item_vocab_gpu_capacity_ratio
250M rows/GPU (1B ÷ 4)
125M rows/GPU (1B ÷ 8)
tested ratio → OOM
tested ratio → FIT
0
50
100
150
200
250
300
350
400
0.0
0.2
0.4
0.6
0.8
1.0